# Cognitive Biases in LLM-Assisted Software Development






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# Even in our minds

2 | 4 | 6

Guess the rule

? | ? | ?

Think of another triple to test your Hypothesis

8 | 10 | 12

“x, x+2, x+4”?

**Fits!**
**H seems right?**

# Even in our minds

| | | | |
|---|---|---|---|
| 2 | 4 | 6 | Given |
| 8 | 10 | 12 | "x, x+2, x+4" Fits! |
| 1 | 10 | 11 | Actual rule: Any Increasing Triple |

# Wason's 2–4–6 rule-discovery task*

Cognitive Bias: systematic deviations from rational judgment that can lead to suboptimal outcome

Maybe you are thinking of x, x+2, x+4 using similar example like?

8 | 10 | 12

Test that confirms the hypothesis

***Confirmation Bias:***
we **tend to** look for evidence that **supports our initial hypothesis**, instead of **evidence that could disprove** it.

But this is a more diagnostic test for the rule x, x+2, x+4:

1 | 10 | 11

Test that challenges the hypothesis

*Wason, P. C. (1960). On the failure to eliminate hypotheses in a conceptual task. Quarterly Journal of Experimental Psychology, 12(3), 129–140. https://doi.org/10.1080/17470216008416717*

# A time of transition!

Traditional Software Development

Developers code by themselves

Cognitive Bias Manifestation well documented

LLM-assited Software Development

92% developers using AI tools

Shift from writing code to evaluating AI generated code

How do biases manifest when working with LLMs?
In this new dynamic of development?

# A time of transition!

Cog


```
// src/Home.js
import React from 'react';
// isUserLoggedIn, userData are passed from the parent component and they come
// from the local storage
const Home = ({ isUserLoggedIn, userData, handleLogIn, handleLogOut }) => {
  const { name, age, email } = userData;

  return (
    <div>
      <h1>Home</h1>
      <p id="greetings">{isUserLoggedIn ? `Welcome ${name}` : 'Please login'}</p>

      {isUserLoggedIn && (
        <div id="user-data">
          <p id="user-data--name">Name: {name}</p>
          <p id="user-data--age">Age: {age}</p>
          <p id="user-data--email">Email: {email}</p>
        </div>
      )}

      {!isUserLoggedIn && (
        <div id="user-login-container">
          <p>Home page is accessible to logged in users only. Please, login:</p>
          <Button
            btnText="Login"
            className="btn btn-primary"
            onClick={handleLogIn}
            type="button"
          />
        </div>
      )}

      {isUserLoggedIn && (
        <Button
          testId="logout-btn"
          btnText="Logout"
          className="btn btn-secondary"
          onClick={handleLogOut}
          type="button"
        />
      )}
    </div>
  )
}

Home.defaultProps = {
  isUserLoggedIn: false,
  userData: {},
  handleLogIn: () => {},
  handleLogOut: () => {}
};

export default Home;
```

# Motivation

Traditional Software Development

Cognitive Bias Manifestation
well documented
Developers code by themselves

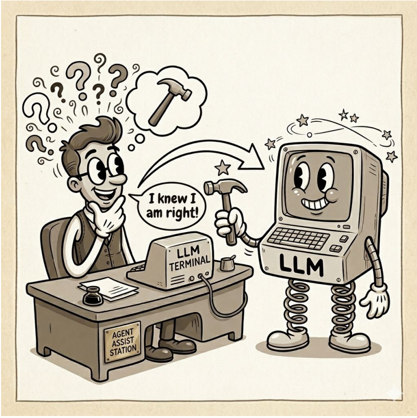
→

LLM-assited Software Development

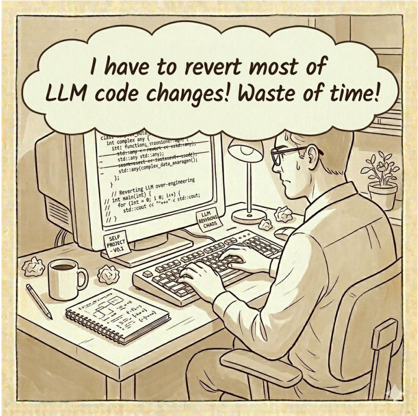
92% developers using AI tools
Shift from writing code to evaluating AI generated code

First Comprehensive Study on Cognitive Biases in LLM-assisted development

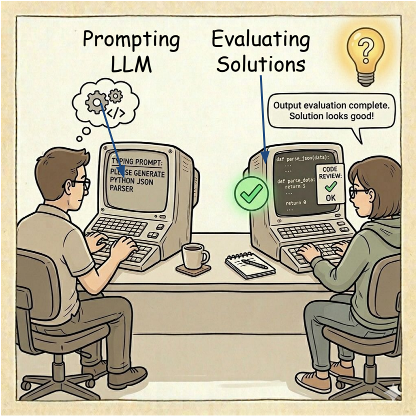
# Research Questions




**RQ1**:
How often do biases appear in LLM-assisted programming?

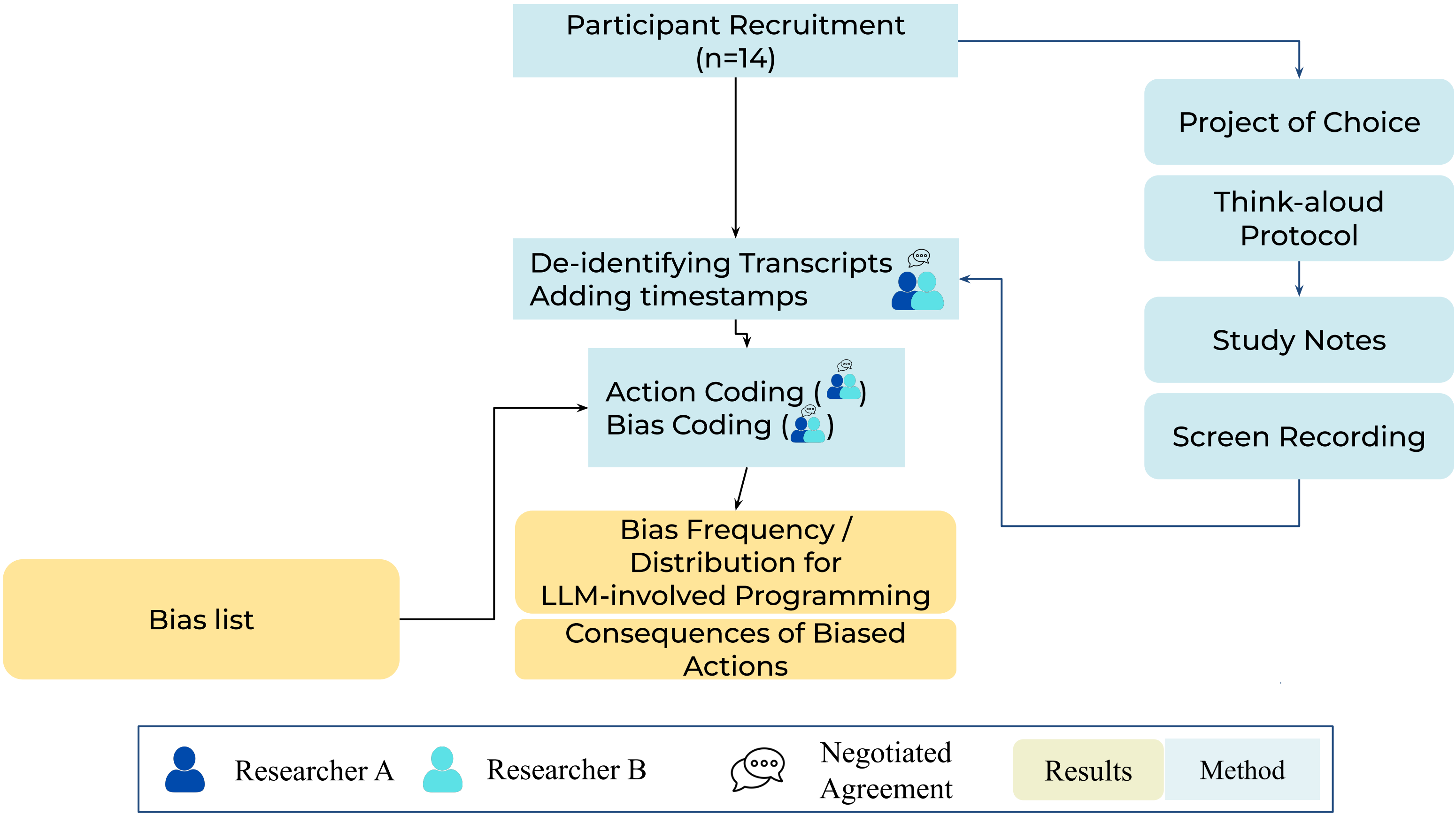



**RQ2**:
What are their impacts?




**RQ3**:
What strategies can help developers mitigate these biases?

# Observational Study



# Measuring impact of bias

Not all biased actions are bad!

| | | |
|---|---|---|
| | Bias | Follow patterns matching bias definition |
| Action | Reversal | Undone, redone, discarded |
| | LLM use | Involve interacting with LLM |

# Observational Study

First, using existing bias list*

| Goal | Timestamp | Action Description | Verbalization | Action Coding | LLM Related | Reversed | Bias Coding |
|---|---|---|---|---|---|---|---|
| Fix a bug | 00:13:59 | Reading error log | Oh here's an error... | Read | ❌ | ✅ | – |
| Copy & pasting LLM codes | 00:14:30 | Adopted LLM response without thinking | I'll just copy and paste it. | AdoptSug | ✅ | ❌ | Instant Gratification |

Transcribing video and audio | Coding Action | Marking Reversals | Coding Bias*

*Chattopadhyay et al. (2020), "A Tale from the Trenches: Cognitive Biases and Software Development," ICSE 2020.

# Chi-square test paradox

$\chi 2$ (1, 2013) = 13.32, $p$ = 0.0003

**High LLM use**

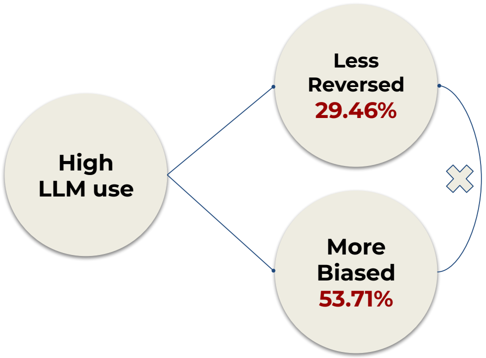
**Reversed**
**29.46%**

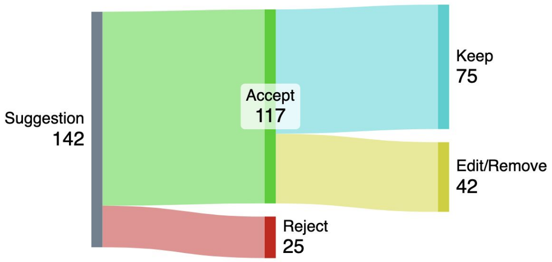
**Biased**
**53.71%**

$\chi 2$ (1, 2013) = 22.94, $p$ = 1.67$e$ −06

*37 Biases Categorized by Previous Work*

# Comparing to Traditional Development




Does LLM generated code evade the critical eye?

Are there new biases? Human-AI? Human-Human?

Or does LLM generated code reduce need for reversal?

*37 Biases Categorized by Previous Work*

*Need New Bias Categorization*

# Bias RE-Categorization

List of Cognitive Bias from Literature Review (239 biases)

↓

Removing duplicates ()
Adding real-life examples ()
Expert Validation ()
Removing Irrelevant Biases ()
Grouping Biases based on previous work ()
Regroup Biases into our categories ()

↓

Final Bias list
(90 biases, 15 categories)

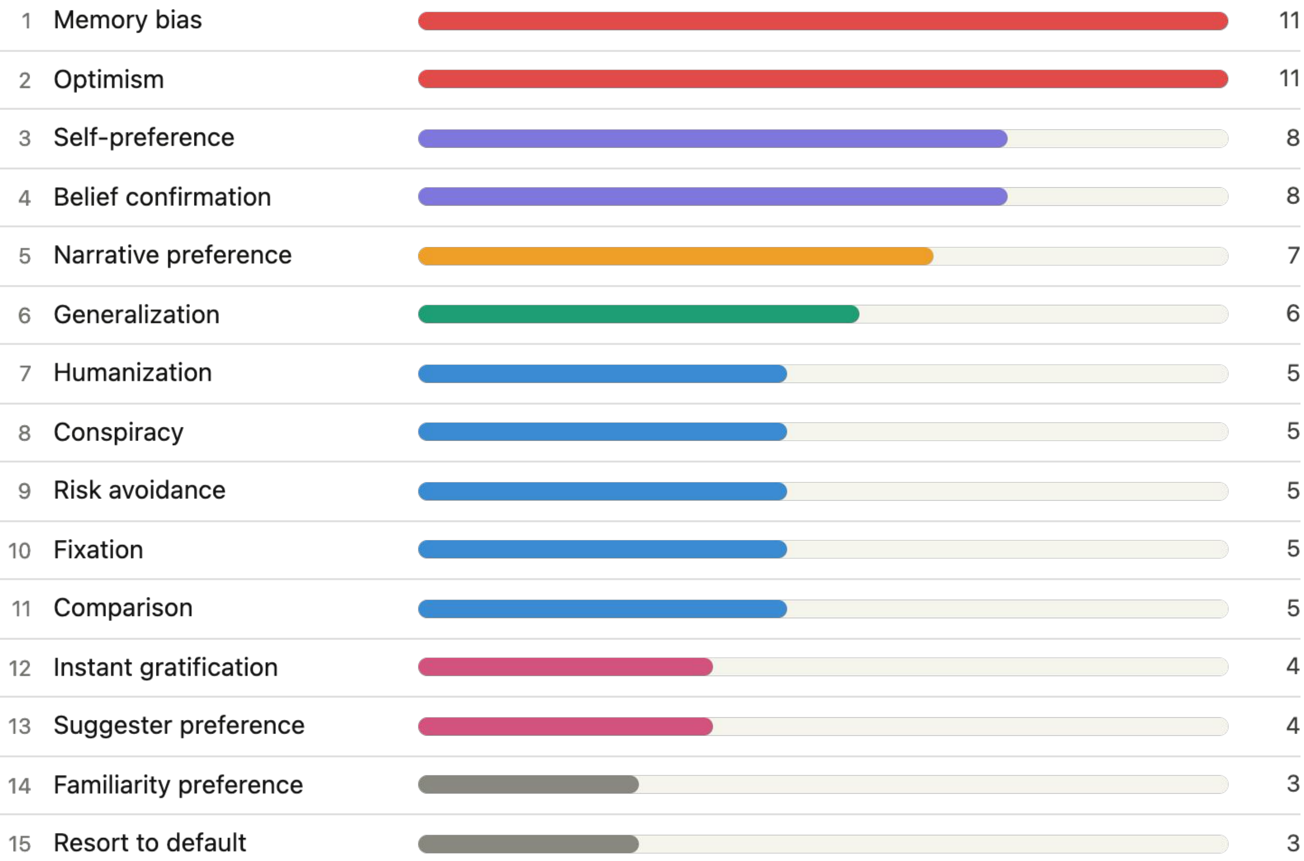
Researcher A | Researcher B | Researcher C | Psychology Expert | Negotiated Agreement | Results | Method

*37 Biases Categorized by Previous Work*

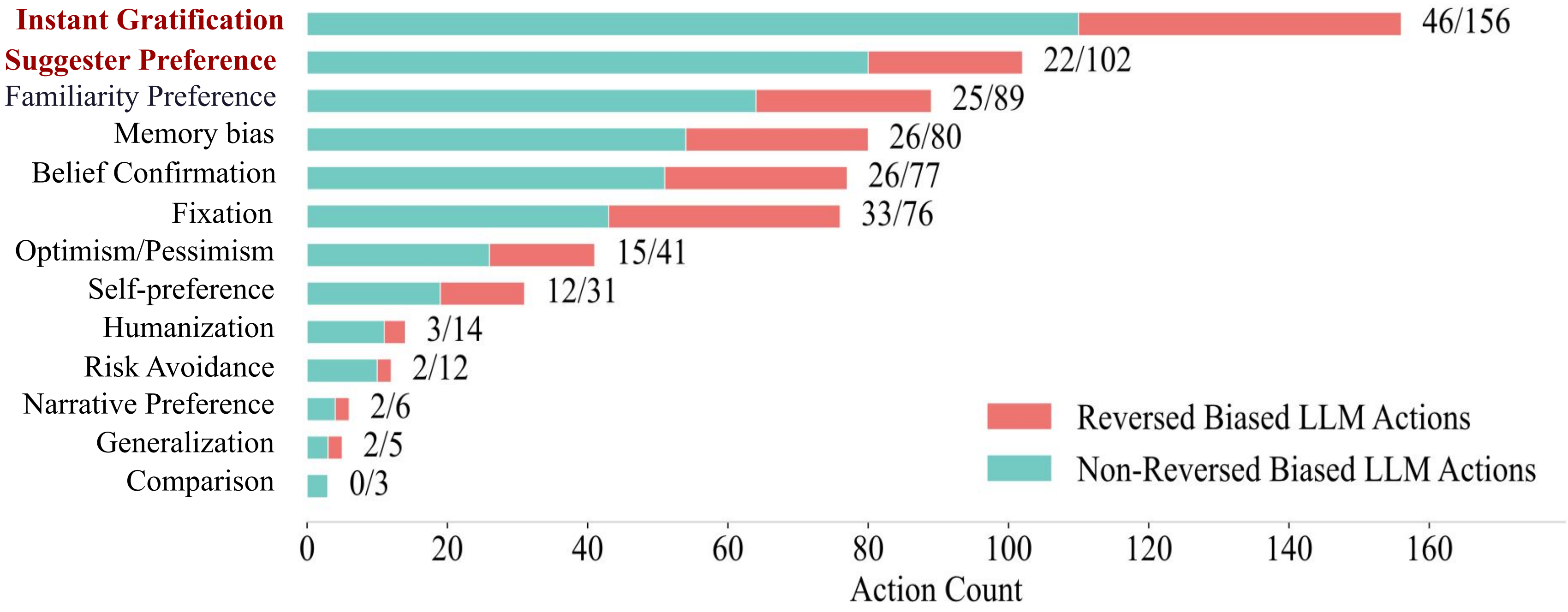
*90 biases 15 categories*

*More interaction based biases*

Number of biases in each category



# Biases Presence and Impacts

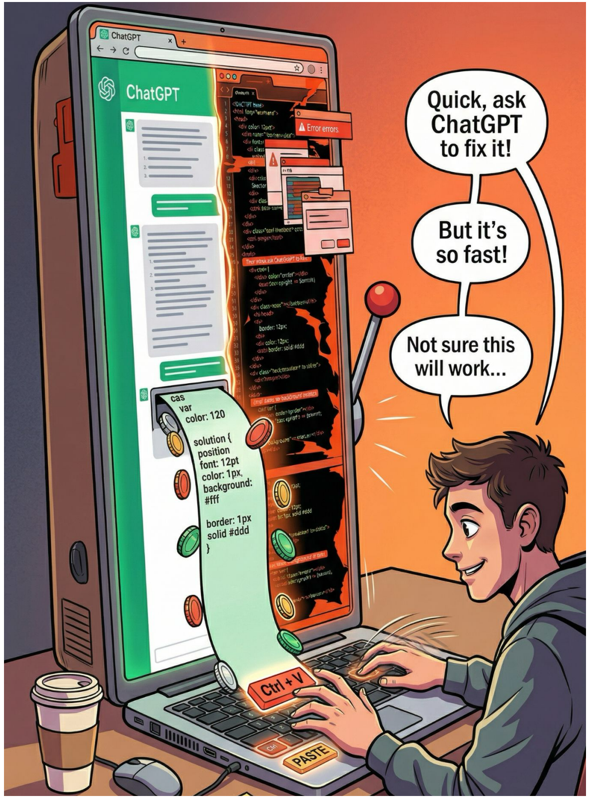


# Biases Presence and Impacts

***Instant Gratification***

P1 copied ChatGPT-generated CSS despite of being ***"not sure it will work"*** which overwrote existing CSS and broke the UI.

Fixing took 17 mins to get back to working state



# Biases Presence and Impacts

*Suggester Preference*

**P13** (Flask dashboard) said he "did not want to deal with HTML" and believed ChatGPT can handle boilerplate

Eventally,
switched to Phind → blind copy+paste → 7 mins

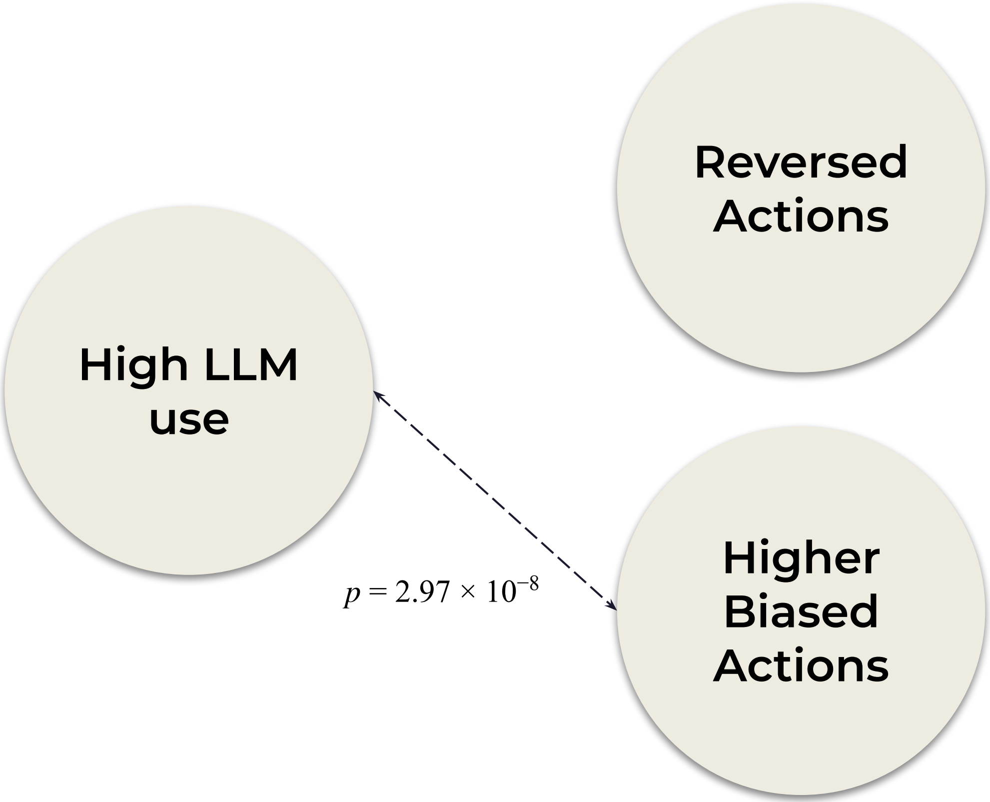
# Biases Presence and Impacts

LLM use

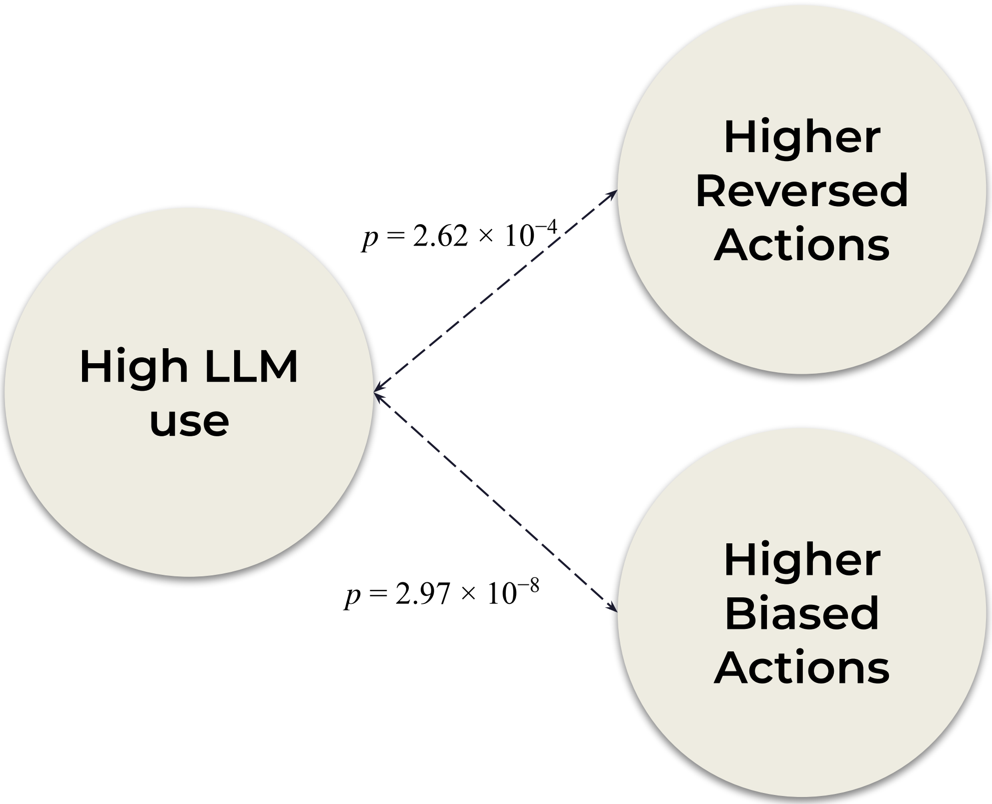
Reversed Actions

Biased Actions

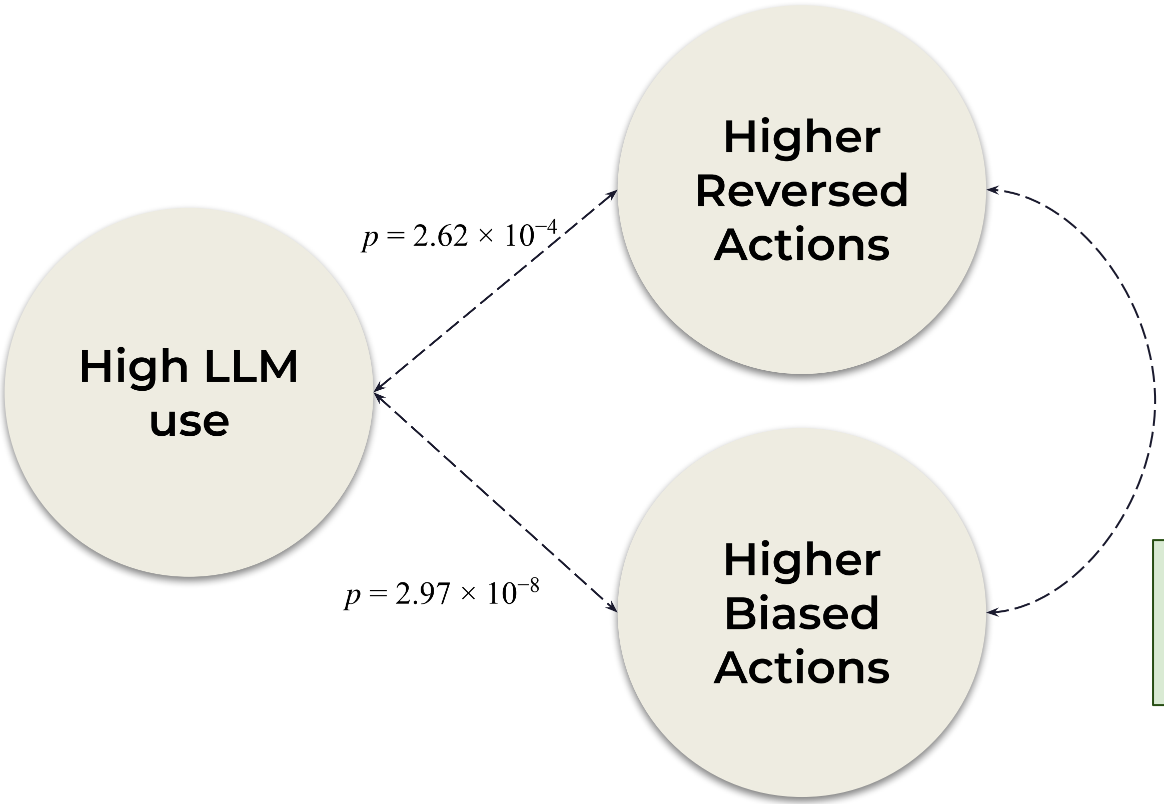
# Chi Square Test



# Chi Square Test



# Chi Square Test

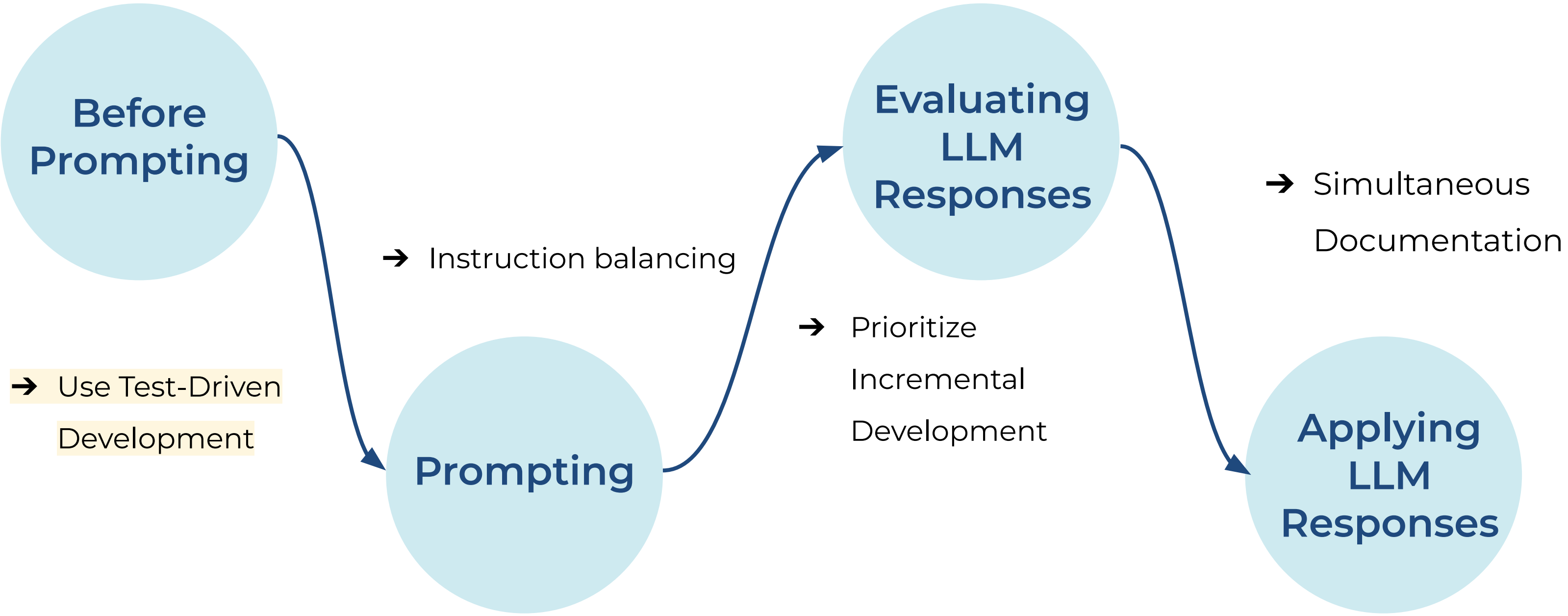



Within LLM interactions:

Fixation ($p = 7.49 \times 10^{-3}$)
Familiarity Bias ($p = 5.67 \times 10^{-4}$)

*LLM-assisted programming increases both bias likelihood and reversal rate.*

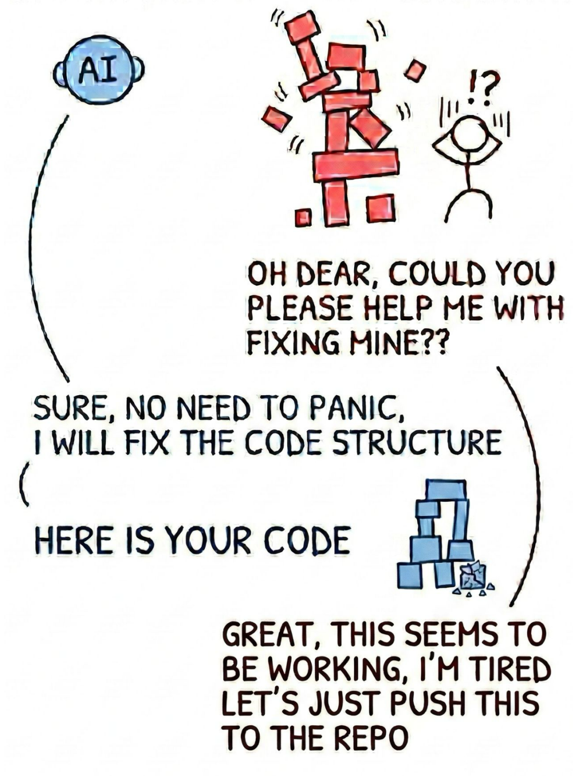
# Helpful Practices



# Implications

- ➔ Illusion of Correctness Issues → Strategic design interventions
- ➔ More Reflective Developer Interaction & Better Decision-Making:

Slow Down the Process → Add friction → Structured Review → System 2 Reasoning



# Thank you for listening!

# Questions & Answers